

HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by background redundancy and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a dual-evidence framework featuring semantic-spectral routing, coverage-constrained spatial optimization, and lightweight downstream detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. To promote recall-safe routing, an object-level soft-coverage loss directly optimizes multi-cell joint instance coverage across ground-truth objects. To recover downstream efficiency, an inverted residual decoupled head (ISPP) is integrated to compress parameters by $> 27\%$ with only fractional FLOPs overhead. Extensive experiments show that HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, establishing an effective accuracy-efficiency co-design suitable for resource-constrained deployment.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ($< 16 \times 16$ px) amid massive uniform backgrounds ($> 70\%$ – 80%) [4, 5]. Evaluating standard dense deep detectors across multi-megapixel grids (1280×1280 to 2048×2048) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* dynamically routes feature processing strictly to prospective foreground regions [4, 6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap directly after a shallow stem, slicing features into candidate bounding patches (AdaSlicer) and bypassing background across downstream stages.

Crucially, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative* across downstream stages [4]. Existing detectors rely exclusively on single semantic objectness with independent per-cell BCE. Under extreme downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we

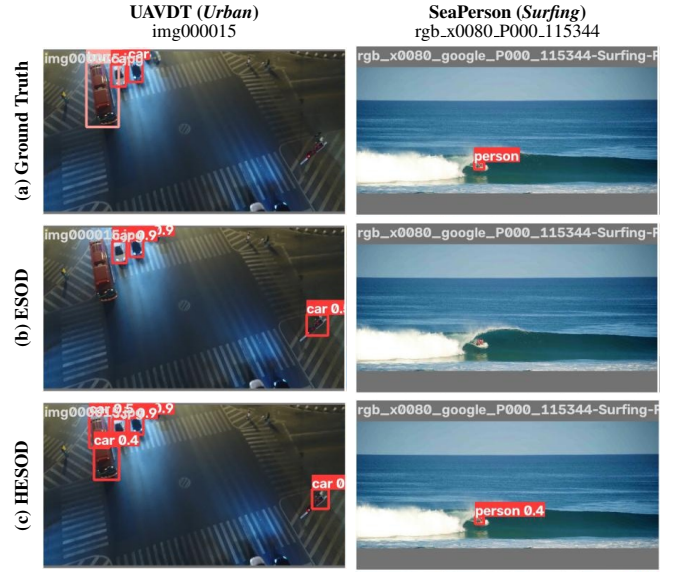


Fig. 1. Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

observe that tiny targets retain prominent high-frequency spatial gradients and spectral saliency even when deep semantics fade [8–10]. We reveal that naive selective inference suffers from acute semantic collapse under asymmetric pruning risks, and propose **HESOD** (High-Resolution Efficient Small Object Detection) to establish recall-safe, evidence-preserving sparse routing. Our key contributions are:

1. **Dual-Evidence Spatial Routing:** We uncover single-semantic routers’ vulnerability to feature collapse, coupling semantic objectness with constant-width gradient filtering to supply scale-robust structural priors independent of backbone capacity.
2. **Evidence Preservation and Soft Coverage:** We formulate a monotonic max-fusion rule preventing multi-cue evidence dilution, paired with an object-level noisy-OR soft-coverage loss directly optimizing joint instance recall under irreversible pruning.
3. **Architectural Compute Compensation:** We co-design downstream detection with an inverted residual decoupled head [11] as a dedicated compute compensator, amortizing candidate evaluation overhead while slashing detector parameters by $> 27\%$.

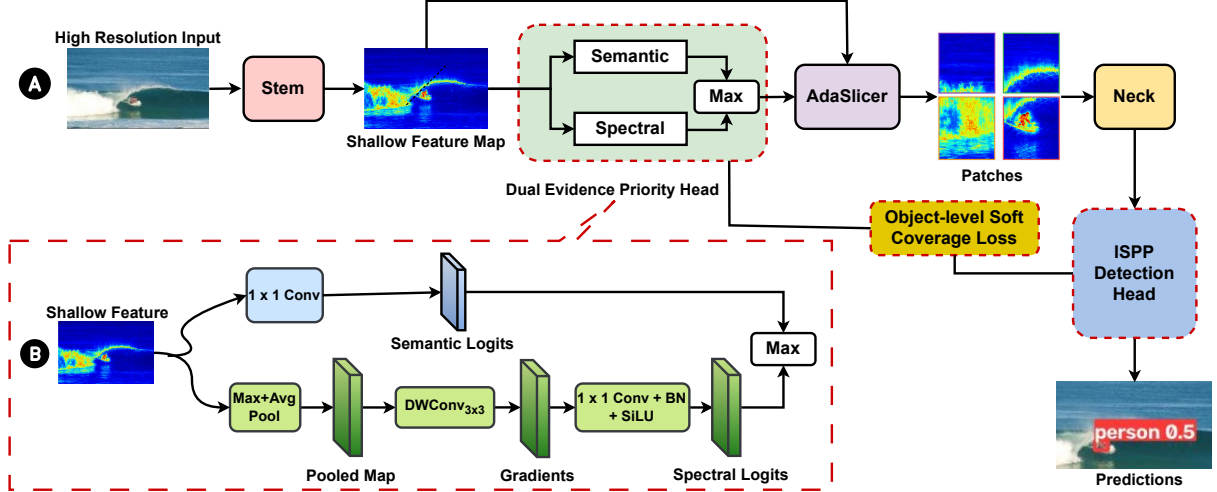


Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Dashed boxes highlight our proposed contributions over ESOD.

2. RELATED WORK

2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ($> 70\%$ – 80%) [1–3, 12]. Classical approaches utilize density-guided cropping (ClusDet [13], DMNet [14]) or scale-adaptive label assignment (NWD [15], RFLA [16]). Recently, ResBiDet [17] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7], and sparse location activation [18]). Recently, coarse-to-fine detectors such as ESOD [4] and OSDT [19] predict early spatial priority heatmaps to extract candidate patches. However, existing selectors rely exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

2.3. Spectral Saliency and Frequency Methods

Frequency-domain representations offer rich structural priors for micro targets, which retain sharp high-frequency edge gradients even when photometric semantics fade [8–10]. While recent methods explore dense frequency transforms [8] or detail compensation across multi-scale feature pyramids [20], they incur substantial downstream latency. In sharp contrast, HESOD repurposes high-frequency gradient saliency upstream as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional

Laplacian and Sobel gradient filtering at constant width, independent of backbone channel dimension C .

3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2(a). From input $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow stem extracts dense features $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer module dynamically clustering candidate foreground cells into bounding patches $\mathcal{P}$ from $\mathbf{F}$. Downstream backbone stages and lightweight decoupled heads evaluate only retained patches $\mathcal{P}$ under soft-coverage supervision (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where $[\cdot \parallel \cdot]$ denotes channel concatenation. Compressing $\mathbf{F}$ to 2 channels retains peak edge activation and background energy with $\mathcal{O}(H_s W_s)$ complexity independent of width C . Next, a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian and directional Sobel operators—filters $\mathbf{F}_{\text{pooled}}$ into 6 structural channels. In the 2D spatial frequency domain, the discrete Laplacian acts as an isotropic high-pass filter with $|H(\omega)| \propto \|\omega\|^2$, naturally amplifying edge transitions. Because micro targets ($< 16^2$ px) manifest as localized high-frequency spatial impulses whose gradient energy resists downsampling [10], this branch supplies a constant-bandwidth prior independent of deep semantic capacity. A 1×1 projection maps features to N_c , yielding spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

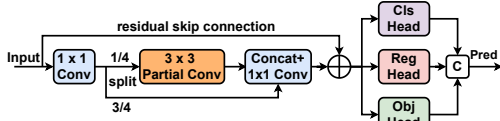


Fig. 3. Inverted Residual Decoupled Head (ISPP) with Partial Convolution (© denotes channel concatenation).

To prevent confidence dilution on faint targets, evidence fusion enforces *evidence preservation*. Linear combinations across heterogeneous evidence spaces are ill-posed because uncalibrated semantic and gradient responses exhibit disparate scales, diluting confident cues. Instead, we formulate a parameter-free elementwise maximum:

$$z_{i,c} = \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}}), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

which guarantees strict monotonic evidence preservation ($z_{i,c} \geq \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}})$) across all cells, functioning as an evidence-preserving OR gate. Candidate cells with routing priority $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$ and forwarded downstream.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. We formulate a noisy-OR differentiable coverage surrogate and its weighted instance loss:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \quad \ell_j = -w_j \log(p_j^{\text{cover}} + \epsilon), \quad (3)$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in cells, penalizing misses on micro targets ($a_j < 4$). For any cell $i \in \mathcal{N}(j)$, the gradient of ℓ_j with respect to cell score s_i is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \quad (4)$$

When neighboring candidate cells have low scores ($s_k \rightarrow 0$), the product term approaches 1, generating a strong gradient to boost cell i and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances: $\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$. The multi-task training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an inverted residual partial-convolution (ISPP) decoupled head [11] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP@.5	Params (M)	GFLOPs	Total Recall	BPR	FPS
UAVDT Benchmark (1280 × 1280)						
Faster R-CNN [21]	0.336	43.3	772.5	79.68%	–	18.8
RetinaNet [22]	0.296	36.4	360.9	83.44%	–	19.8
QueryDet [†] [6]	0.257	39.3	– [†]	41.89%	–	21.1
ESOD [4]	<u>0.385</u>	<u>35.9</u>	68.2	85.17%	<u>0.884</u>	117.8
HESOD (Ours)	0.394	26.0	<u>74.9</u>	88.83%	0.940	<u>106.1</u>
Δ vs. ESOD	+0.9 pp	-27.5%	+9.8%	+3.66 pp	+5.6 pp	-9.9%
SeaPerson Benchmark (2048 × 2048)						
Faster R-CNN [21]	0.551	43.3	1546.8	67.27%	–	23.1
RetinaNet [22]	0.473	36.4	942.7	65.98%	–	28.0
QueryDet [†] [6]	0.627	39.3	– [†]	61.56%	–	14.0
ESOD [4]	<u>0.750</u>	<u>35.8</u>	202.4	84.42%	<u>0.947</u>	85.7
HESOD (Ours)	0.773	25.9	<u>208.5</u>	87.75%	0.991	<u>80.6</u>
Δ vs. ESOD	+2.3 pp	-27.6%	+3.0%	+3.33 pp	+4.4 pp	-6.0%

BPR: Bounding Patch Recall. **Bold**: best; underline: second-best (selective models).

[†]QueryDet uses CSQ (dynamic sparse FLOPs unamenable to static profiling; dense query counterparts: 973.5 / 2539.0 GFLOPs at 12.5 / 10.6 FPS on UAVDT / SeaPerson).

Two-Stage Training Protocol. To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1 (Selector Convergence)*: Train the Dual-Max selector with a coupled head under $\mathcal{L}_{\text{cover}}$, establishing a well-calibrated priority heatmap; and (ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak coverage (0.940 BPR), maintaining competitive real-time FPS with only fractional overhead vs. ESOD.

4. EXPERIMENTS

4.1. Experiment Settings

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280 × 1280 px, urban vehicle surveillance across 3 classes) and **SeaPerson** [3, 12] (2048 × 2048 px, maritime search-and-rescue with $> 85\%$ micro targets < 32 px). Both evaluate full-resolution images without artificial cropping.

All models employ YOLOv5m in PyTorch 2.8 and train for 50 epochs under selective protocols [4]; HESOD trains the selector in Stage 1 and fine-tunes the lightweight ISPP head with the frozen selector in Stage 2 to maintain routing stability. Metrics include COCO mAP@.5, scale-stratified recall across bins (*Very Tiny* $< 16^2$, *Tiny* 16^2 – 32^2 , *Small* 32^2 – 96^2), and *Bounding Patch Recall* (BPR = $N_{\text{cov}}/N_{\text{gt}}$, where a target is covered if $|\text{GT} \cap \mathcal{P}|/|\text{GT}| > 0.5$ by selected patches $\mathcal{P}$). To guarantee strict comparability, latency (FPS) across all baselines and our models are re-benchmarked on NVIDIA RTX 5090 GPU.

4.2. Main Benchmark Comparisons

Comparison with Dense and Sparse Baselines. As shown in Table 1, standard detectors incur prohibitive latency on multi-megapixel grids. On **UAVDT**, HESOD outperforms Faster R-CNN, RetinaNet, and QueryDet [6] by **+5.8 pp**, **+9.8 pp**, and **+13.7 pp** mAP@.5 at **5.0×–5.6×** higher FPS (106.1 vs. 18.8–21.1) with higher recall (88.83% vs. 41.89%–83.44%). On **SeaPerson**, HESOD surpasses all baselines by **+14.6~+30.0 pp** mAP@.5 at **2.9×–5.8×** higher FPS (80.6 vs. 14.0–28.0), lifting total recall to

Table 2. Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

Configuration	Detection Accuracy and Computational Efficiency						Physical-Scale Recall Breakdown				
	mAP @.5	mAP @.5:.95	BPR	Params (M)	GFLOPs	FPS	Very Tiny ($< 16^2$)	Tiny (16^2-32^2)	Small (32^2-96^2)	Med/Lrg ($> 96^2$)	Total Recall
UAVDT Benchmark (1280×1280)							(74,375)	(206,015)	(86,282)	(7,325)	(373,997)
(1) ESOD Baseline*	0.385	0.214	0.884	35.9	68.2	117.8	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	<u>0.217</u>	0.906	35.9	75.0	<u>113.8</u>	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	0.387	0.209	<u>0.942</u>	36.0	93.7	102.3	82.43%	88.72%	96.66%	<u>66.80%</u>	88.87%
(4) (3) + Channel-Pooled	0.394	0.209	0.963	35.9	99.3	97.6	86.88%	92.23%	97.74%	66.43%	91.93%
(5) Dual-Concat	0.371	0.205	0.919	35.9	83.9	107.2	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	0.395	0.218	0.940	35.9	90.1	102.3	<u>85.69%</u>	<u>89.97%</u>	<u>97.37%</u>	65.99%	<u>90.36%</u>
(7) HESOD (Ours)	0.394	0.215	0.940	26.0	<u>74.9</u>	106.1	80.76%	88.98%	97.11%	69.16%	88.83%
SeaPerson Benchmark (2048×2048)							(82,417)	(174,816)	(42,987)	(155)	(300,375)
(1) ESOD Baseline*	0.750	0.320	0.947	35.8	202.4	85.7	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	0.325	0.991	35.8	266.8	73.5	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	<u>0.327</u>	0.992	35.8	267.4	71.7	75.23%	<u>91.69%</u>	95.89%	<u>86.45%</u>	87.77%
(4) (3) + Channel-Pooled	0.767	0.324	0.988	35.8	263.4	72.7	76.13%	91.36%	95.50%	87.10%	87.77%
(5) Dual-Concat	0.772	0.326	0.986	35.8	281.2	69.8	<u>76.00%</u>	91.50%	95.68%	80.00%	<u>87.84%</u>
(6) Dual-Max	0.778	0.330	<u>0.991</u>	35.8	255.1	73.8	75.65%	92.06%	<u>95.85%</u>	81.94%	88.10%
(7) HESOD (Ours)	<u>0.773</u>	<u>0.327</u>	<u>0.991</u>	25.9	<u>208.5</u>	<u>80.6</u>	75.71%	91.47%	95.72%	80.65%	87.75%

*Trained with standard BCE (others use $\mathcal{L}_{\text{cover}}$). Parenthetical values denote ground-truth counts per bin. One-to-one matching at $\text{conf} \geq 0.001$, $\text{IoU} \geq 0.5$. **Bold:** best; underline: second-best.

87.75% (vs. 61.56%–67.27%; on Very Tiny: 48.73% $\rightarrow$ **75.71%**, +27.0 pp over QueryDet), confirming selective routing efficacy.

Comparison with Selective Baseline (ESOD). Against ESOD, HESOD achieves substantial recall recovery and parameter compression under strict recall-safe efficiency (Table 1). Because selective models evaluate only routed regions, baseline ESOD achieves lower FLOPs only by prematurely discarding micro targets (HESOD rescues $> 13,700$ targets on UAVDT). Forwarding these genuine candidate regions downstream naturally consumes modest compute. On **SeaPerson** ($> 85\%$ targets < 32 px), HESOD boosts mAP@.5 by +2.3 pp (0.750 $\rightarrow$ **0.773**) and BPR by +4.4 pp to **0.991** (total recall +3.33 pp to **87.75%**), while cutting parameters by **27.6%** at 80.6 FPS with only +3.0% net GFLOPs. On **UAVDT**, HESOD improves mAP@.5 to **0.394** (+0.9 pp) and BPR to **0.940** (+5.6 pp), cutting parameters by **27.5%** at 106.1 FPS with marked recall gains across *car* (+3.4 pp), *truck* (+5.4 pp), and *bus* (67.7% $\rightarrow$ 81.4%, +13.7 pp).

4.3. Ablation Studies and Scale Analysis

Evidence Preservation Principle. As reported in Table 2, unconstrained affine concatenation (Arm 5, 0.371 mAP) degrades performance below single-evidence branches (0.384 and 0.387), confirming that linear projections across heterogeneous evidence spaces dilute confident signals due to disparate dynamic ranges. In sharp contrast, parameter-free Elementwise Max (Arm 6) enforces strict monotonic evidence preservation ($z_{i,c} \geq \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}})$) as an evidence-preserving OR gate, achieving peak selector accuracy of **0.395** mAP on UAVDT and **0.778** mAP on SeaPerson (0.991 BPR) with stable high-IoU localization (mAP@.5 : .95 of 0.218 and 0.330).

Efficiency Recovery via Lightweight Head. The inverted residual ISPP head serves as a dedicated compute compensator to amortize the arithmetic cost of evaluating rescued candidate regions. Without ISPP, uncompressed Dual-Max incurs 90.1 GFLOPs on UAVDT and 255.1 GFLOPs on SeaPerson. On UAVDT, integrating ISPP (Arm 7) preserves 0.394 mAP@.5 and 0.940 BPR while slashing GFLOPs by 16.9% (90.1 $\rightarrow$ 74.9) and parameters by **27.5%** at 106.1 FPS. On SeaPerson, ISPP slashes GFLOPs by **18.2%** (255.1 $\rightarrow$ 208.5) and parameters by **27.6%** with BPR fully preserved at 0.991, raising FPS from 73.8 to 80.6 with negligible accuracy cost (0.778 $\rightarrow$ 0.773 mAP).

Micro-Scale Recall Recovery. In Table 2, baseline ESOD suffers severe degradation on *Very Tiny* targets ($< 16^2$ px), dropping to 79.43% on UAVDT and 74.03% on SeaPerson due to early semantic collapse. Dual-Max routing restores extreme-scale sensitivity across both domains, boosting *Very Tiny* recall by +6.26 pp on UAVDT and improving *Tiny* recall by +5.28 pp on SeaPerson (92.06% vs. 86.78%, with *Very Tiny* recall reaching **75.71%**).

Channel Pooling Benefit. Crucially, channel pooling resolves memory bottlenecks in the Dual-Max selector. On SeaPerson, full-width spectral filtering causes severe memory pressure (limiting batch size to 2) and yields only 0.763 mAP@.5 at 277.8 GFLOPs. Compressing features to 2 channels enables batch-8 training, lifting mAP@.5 by +1.5 pp (0.763 $\rightarrow$ **0.778**), boosting BPR to **0.991**, and cutting GFLOPs by **8.9%**. Diagnostics (Arms 3 vs. 4) confirm that channel pooling suppresses noise and lifts *Very Tiny* recall on both benchmarks (+4.45 pp on UAVDT, +0.90 pp on SeaPerson).

Error Bottleneck Diagnosis. Dual-evidence routing and coverage optimization slash selector false negatives from 25.6% to 19.2% on SeaPerson (**99.1%** BPR) and rescue $> 13,700$ dropped instances on UAVDT (BPR 0.884 $\rightarrow$ **0.940**), shifting the remaining bottleneck to downstream regression refinement.

Latency and Qualitative Profiling. On an RTX 5090 GPU, stem extraction takes 1.2 ms, selector 1.8 ms, AdaSlicer 0.9 ms, and ISPP 5.5 ms (106.1/80.6 FPS on UAVDT/SeaPerson). As visualized in Fig. 1, HESOD reliably detects faint micro targets missed by ESOD without generating false alarms.

5. CONCLUSION

We presented HESOD, an efficient dual-evidence framework for high-resolution small object detection. Uniting semantic-spectral routing, monotonic evidence preservation, and soft coverage ($\mathcal{L}_{\text{cover}}$), HESOD prevents premature pruning of micro targets. Coupled with an ISPP decoupled head, HESOD boosts mAP@.5 by 0.9 pp and 2.3 pp (+3.66 pp and +3.33 pp total recall) on UAVDT and SeaPerson while compressing parameters by $> 27\%$, achieving a practical accuracy-efficiency co-design for resource-constrained deployment.

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